Show Word Count : Yes
Answers Type : Set
Answers Case Sensitive : No
Text Areas : PlainText
Possible Answers :

g,r,b

g,b,r

Deep Learning

Section Id :	64065399901
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	10
Number of Questions to be attempted :	10
Section Marks :	40
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653230714
Question Shuffling Allowed :	No

Question Number : 97 Question Id : 6406531425468 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : DEEP LEARNING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534764441. ✔

YES

6406534764442. ✖

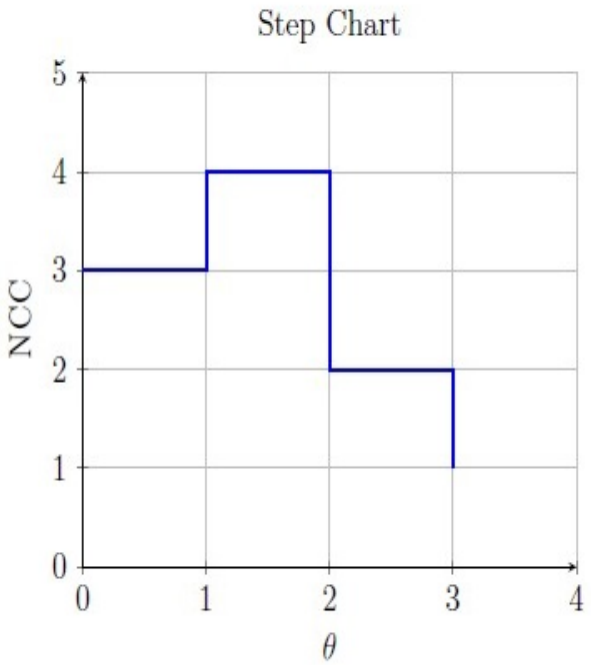
NO

Sub-Section Number :	2
Sub-Section Id :	640653230715
Question Shuffling Allowed :	Yes

Question Number : 98 Question Id : 6406531425469 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Suppose that we implement a two input Boolean function using the McCulloch Pitts (MP) neuron. The graph below shows the Number of Correctly Classified (NCC) data points for various values of threshold θ. The threshold θ is incremented by 1 from 0 to 3. Assume that the neuron does not have any inhibitory input.



This graph represents which of the following Boolean functions?

Options :

6406534764443. ✖

XOR

6406534764444. ✖

AND

6406534764445. ✔

OR

6406534764446. ✖

NAND

6406534764447. ✖

None of these

Question Number : 99 Question Id : 6406531425470 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Based on the two statements provided choose the correct option?

Statement 1: Optimization's primary goal is to reduce the training error.

Statement 2: Regularization helps to prevent overfitting, which in turn reduces the training error.

Options :

6406534764448. ✖

Both Statement 1 and Statement 2 are false.

6406534764449. ✔

Statement 1 is true, but Statement 2 is false.

6406534764450. ✖

Statement 1 is false, but Statement 2 is true.

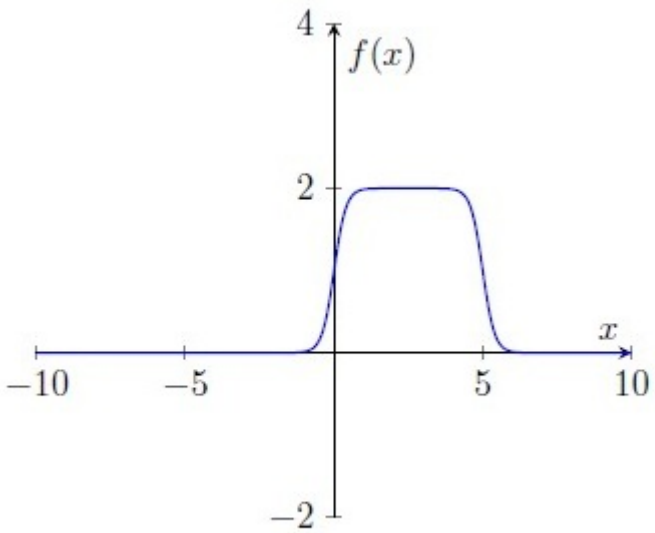
6406534764451. ✖

Both Statement 1 and Statement 2 are true.

Question Number : 100 Question Id : 6406531425471 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Consider the tower function plot shown below:



Which of the following could be the correct equation for this plot?

Options :

6406534764452. ✖ $f(x) = \frac{1}{1+e^{-5x+25}} - \frac{1}{1+e^{-5x}}$

6406534764453. ✔ $f(x) = \frac{2}{1+e^{-5x}} - \frac{2}{1+e^{-5x+25}}$

6406534764454. ✖ $f(x) = \frac{2}{1+e^{-5x+25}} - \frac{2}{1+e^{-5x}}$

6406534764455. ✖ $f(x) = \frac{1}{1+e^{-5x}} - \frac{1}{1+e^{-5x+25}}$

Sub-Section Number : 3

Sub-Section Id : 640653230716

Question Shuffling Allowed : Yes

Question Number : 101 Question Id : 6406531425472 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a feedforward neural network with one hidden layer trained using backpropagation for a binary classification task. The network has the following architecture:

- Input layer with 15 neurons
- Hidden layer with 25 neurons
- Output layer with 1 neuron

Assume that:

- The activation function for both the hidden and output layers is the sigmoid function, $\sigma(z)$,
- The derivative of the sigmoid function is $\sigma'(z) = \sigma(z)(1 - \sigma(z))$,
- The loss function used is binary cross-entropy (BCE),
- $\hat{y}^{(i)}$ is the output for the i-th example, $y^{(i)}$ is the true label, and $h_j^{(i)}$ is the output of the j-th hidden neuron for the i-th example,
- There are no biases in the network.

Let w_j represent the weight connecting the jth neuron of the hidden layer to the output neuron. Identify the **correct weight update rule** Δw_j for each type of gradient descent: Vanilla Gradient Descent (GD) ,mini-Batch Gradient Descent (BGD), and Stochastic Gradient Descent (SGD). (Recall: with BCE + sigmoid, the gradient w.r.t. the pre-activation simplifies to $\hat{y}^{(i)} - y^{(i)}$.

Options :

6406534764456. ✔ GD: $\Delta w_j = -\eta \cdot \frac{1}{N} \sum_{i=1}^{N} \left(\hat{y}^{(i)} - y^{(i)}\right) h_j^{(i)}$

6406534764457. ✔ BGD: $\Delta w_j = -\eta \cdot \frac{1}{B} \sum_{i=1}^{B} \left(\hat{y}^{(i)} - y^{(i)}\right) h_j^{(i)}$

6406534764458. ✔ SGD: $\Delta w_j = -\eta \cdot \left(\hat{y}^{(i)} - y^{(i)}\right) h_j^{(i)}$

6406534764459. ✖ GD: $\Delta w_j = -\eta \cdot \frac{1}{N} \sum_{i=1}^{N} \left(\hat{y}^{(i)} - y^{(i)}\right) \sigma'\left(\sum_{k=1}^{25} w_k h_k^{(i)}\right) h_j^{(i)}$

6406534764460. ✖ BGD: $\Delta w_j = -\eta \cdot \frac{B}{N} \sum_{i=1}^{N} \left(\hat{y}^{(i)} - y^{(i)}\right) h_j^{(i)}$

6406534764461. ✖ SGD: $\Delta w_j = -\eta \cdot \frac{1}{B} \sum_{i=1}^{B} \left(\hat{y}^{(i)} - y^{(i)}\right) h_j^{(i)}$

Sub-Section Number :	4
Sub-Section Id :	640653230717
Question Shuffling Allowed :	No

Question Id : 6406531425473 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (102 to 104)

Question Label : Comprehension

Scaled Dot-Product Attention

You are given a simple transformer attention setup with the following configuration:

- Embedding dimension: $d_{\text{model}} = 2$
- Input sequence (sequence length = 2, each column vector in $\mathbb{R}^2$):

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

- Dimension of queries and keys: $d_k = 2$
- Projection matrices:

$$W_Q = W_K = \begin{bmatrix} 2 & -1 \\ 1 & -1 \end{bmatrix}, \quad W_V = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$$

Scaled Dot-Product Attention:

$$\text{Scaled Dot-Product Attention}(Q, K, V) = \text{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V^T$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 102 Question Id : 6406531425474 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Compute the matrix $Q^T K$. What is the **sum of all values** in this matrix?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10

Question Number : 103 Question Id : 6406531425475 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Compute the scaled matrix $\frac{Q^T K}{\sqrt{d_k}}$, then apply **row-wise softmax**. What is the **sum of the values in the last column** of the resulting matrix? (Answer up to **2 decimal places**.)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.52 to 1.60

Question Number : 104 Question Id : 6406531425476 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Compute the final attention output:

$$\text{Scaled Dot-Product Attention}(Q, K, V) = \text{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V^T$$

What is the **sum of the values in the first row** of the Scaled Dot-Product Attention output?
(Answer up to **2 decimal places**.)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :

3.95 to 4.03

Sub-Section Number :	5
Sub-Section Id :	640653230718
Question Shuffling Allowed :	No

Question Id : 6406531425477 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (105 to 107)

Question Label : Comprehension

Consider an image X of size 4×4, and a true label $y = 3$. The image is passed through a convolutional neural network with a single filter followed by a ReLU and a max-pooling operation.

$$X = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 \\ 1 & 0 & 2 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Parameters:

- Convolution: stride $S = 2$, padding $P = 1$
- Max pooling: filter size $F = 2$, stride $S = 2$, padding $P = 0$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 105 Question Id : 6406531425478 Question Type : MCQ Correct Marks : 2

Question Label : Multiple Choice Question

What is the output of the convolution layer after applying ReLU?

Options :

6406534764465. ✖ $\begin{bmatrix} 5 & 3 \\ 4 & 4 \end{bmatrix}$

6406534764466. ✔ $\begin{bmatrix} 2 & 6 \\ 3 & 5 \end{bmatrix}$

6406534764467. ✖ $\begin{bmatrix} 4 & 5 \\ 5 & 4 \end{bmatrix}$

6406534764468. ✖ $\begin{bmatrix} 7 & 4 \\ 4 & 4 \end{bmatrix}$

Question Number : 106 Question Id : 6406531425479 Question Type : SA
Correct Marks : 1
Question Label : Short Answer Question

What is the predicted output $\hat{y}$ after max-pooling?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
6

Question Number : 107 Question Id : 6406531425480 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

Given loss $L(y, \hat{y}) = \frac{1}{2}(\hat{y} - y)^2$, compute $\frac{\partial L}{\partial K_{33}}$ where K_{33} is the bottom-right element of the convolution filter.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Possible Answers :

6

Sub-Section Number : 6

Sub-Section Id : 640653230719

Question Shuffling Allowed : No

Question Id : 6406531425481 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (108 to 110)

Question Label : Comprehension

Consider a CBOW model for learning word embeddings. The vocabulary is made up of three words, {good, bad, ugly}. W and C are the matrices that contain the word and context embeddings respectively. The columns in each matrix correspond to the embeddings. Both matrices are of shape 2×3:

$$W = \begin{bmatrix} \text{good} & \text{bad} & \text{ugly} \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}, \; C = \begin{bmatrix} \text{good} & \text{bad} & \text{ugly} \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix}$$

The context window is 1, meaning, the next word is predicted using just the current word as context. Recall that we use softmax to make predictions at the output.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 108 Question Id : 6406531425482 Question Type : SA Correct Marks : 2

Question Label : Short Answer Question

With what probability does the model output the word "bad" given the word "good" as context? In other words, find $P(\text{bad} \mid \text{good})$. Enter your answer correct to two places after the decimal.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

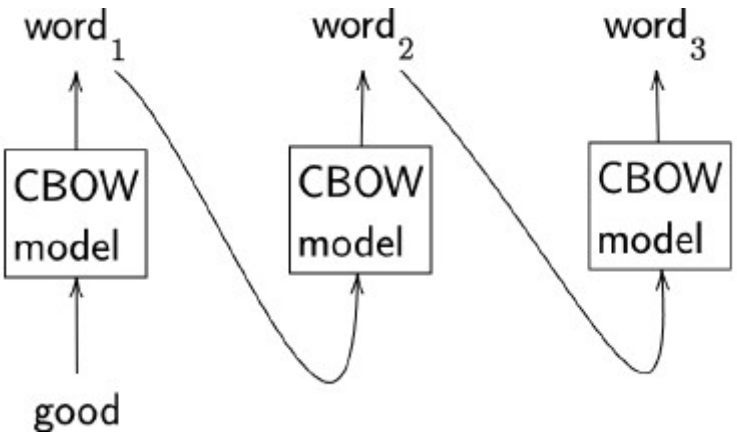
0.62 to 72

Question Number : 109 Question Id : 6406531425483 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

The CBOW model is now used to generate a “sentence” or a string of words. First we pass the word “good” and retain the word with highest probability as the output, say $word_1$, which is in turn passed as input to the model. If the model is run this way for exactly three time steps, what is the sentence that it outputs? Note that the sentence here is “$word_1$ $word_2$ $word_3$”.



Options :

6406534764472. ✔

bad ugly good

6406534764473. ✖

good bad ugly

6406534764474. ✖

bad bad bad

6406534764475. ✖

good good good

6406534764476. ✖

bad ugly bad

Question Number : 110 Question Id : 6406531425484 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Now consider updating the word embeddings using the sample “good good”. The first “good” in the string is used as context and the second “good” as the true label. Use cross entropy as the loss function and run one iteration of gradient descent with $\eta = 1$ starting with the existing values for the embeddings.

Find the updated word embedding for “good” and choose the most appropriate option from below. Note that you have to compute the updated word embedding for “good” and not its context embedding.

Options :

6406534764477. ✔

(1.76, -0.24)

6406534764478. ✖

(1.24, -0.76)

6406534764479. ✖

(1.76, 1.76)

6406534764480. ✖

(1.76, -1.76)

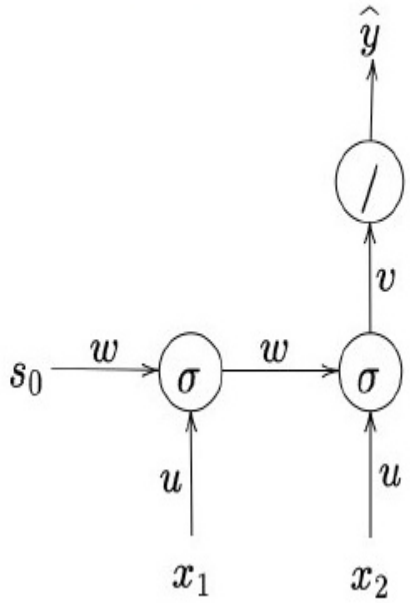
6406534764481. ✖

(1.24, -1.24)

Question Id : 6406531425485 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (111 to 114)

Question Label : Comprehension

Consider a toy-RNN for a regression problem with input $\in \mathbb{R}$ and just one neuron in the hidden layer unrolled in time for two time steps.



The hidden layer neuron is a sigmoid neuron and the output layer neuron is linear.

- w is the weight corresponding to the recurrent connection
- u is the weight of the connection between the input and the hidden layer neuron
- v is the weight of the connection between the hidden layer neuron and the output neuron
- s_0 is the hidden state at time step $t = 0$. s_1 and s_2 are the outputs (activations) of the hidden layer neurons at time step $t = 1$ and $t = 2$ respectively.

Ignore biases everywhere. The loss is squared error and given by $L(y, \hat{y}) = \frac{1}{2} \cdot (\hat{y} - y)^2$.

Numerical Data

- $w = 1,\ u = 1,\ v = 2$
- $s_0 = 0$
- $x_1 = -\log_e 3,\ x_2 = \frac{-1}{4}$ and $y = 0$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 111 Question Id : 6406531425486 Question Type : SA
Correct Marks : 1

Question Label : Short Answer Question

Find s1.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes

Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.25

Question Number : 112 Question Id : 6406531425487 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Find s2.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.5

Question Number : 113 Question Id : 6406531425488 Question Type : SA
Correct Marks : 1
Question Label : Short Answer Question
Find $\hat{y}$.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 114 Question Id : 6406531425489 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
Find $\frac{\partial L}{\partial w}$.
Hint: Expressing $\hat{y}$ as a function of u, v, w, and then computing the partial derivative might help.

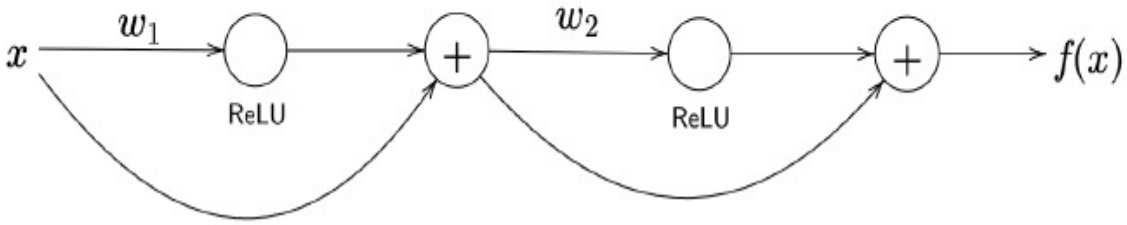
Response Type : Numeric
Evaluation Required For SA : Yes

Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.125

Sub-Section Number :	7
Sub-Section Id :	640653230720
Question Shuffling Allowed :	No

Question Id : 6406531425490 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (115 to 116)
Question Label : Comprehension

Consider a setup where we use what are called skip connections. The operation $\oplus$ adds the output activation of the ReLU neuron before it and the input to that ReLU neuron. For example, the first $\oplus$ would output $x + \max(0, w_1 x)$. Ignore biases.



Based on the above data, answer the given subquestions.
Sub questions

Question Number : 115 Question Id : 6406531425491 Question Type : SA Correct Marks : 2
Question Label : Short Answer Question

If $w_1 = 1, w_2 = -5$, compute $f(1)$.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
2

Question Number : 116 Question Id : 6406531425492 Question Type : SA Correct Marks : 2
Question Label : Short Answer Question